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Research on the Adaptability of SP Flooding in Sand-Gravel Mixture Reservoir Based on the Inverted Seven-Spot Well Pattern

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Abstract

As one of the leading technologies for chemical enhanced oil recovery (cEOR), surfactant-polymer (SP) flooding technology has drawn the attention of petroleum scientists and engineers for many years. However, most of its application scenarios are based on the five-spot well pattern. Rarely reported is its EOR potential in an inverted seven-spot well pattern. Based on the physical properties of Karamay Oilfield in China, this paper studied the adaptability of the SP system in the inverted seven-spot well pattern. Firstly, the numerical simulation method and the single-core flooding experiment were used to compare the seepage intensities of the two well patterns and the EOR ability of the SP system under different seepage intensities. Then, the migration law and the oil displacement effect of the SP system under the conditions of sand-gravel mixture were evaluated. Finally, the EOR ability under different injection strategies in the well patterns was evaluated. The results show that the inverted seven-spot well pattern shows a weak swept state, accounting for 61% of the whole region. Appropriately increasing the viscosity and slug size of the SP system improves the oil production of the low-permeability conglomerate layer. Step-down viscosity injection can further enlarge the sweep range of injection fluid and enhance oil recovery compared to constant viscosity injection. Compared with the five-spot well pattern, the swept area of the SP system in the inverted seven-spot well pattern is larger while the strength is weaker. The injection and production wells should be reasonably arranged when the well pattern is converted to efficiently recover the remaining oil and residual oil that are not recovered in the five-spot well pattern by utilizing the inverted seven-spot well pattern characteristics.

Introduction

Due to exponential population growth and rapid industrialization, crude oil is deemed the primary energy source for humankind. Unless other alternatives and economical methods of energy generation are discovered, it will continue to dominate the energy market in the near future. Compared with other mineral resources, oil recovery is relatively low. The oil production obtained by relying solely on natural energy (primary oil recovery) and water injection (secondary oil recovery) is limited. In such cases, a substantial portion (about 65% – 77%) of the remaining oil is still confined in the reservoirs (Tang et al., 2018, 2020).

The SP (surfactant-polymer) flooding technology is a kind of tertiary oil recovery technologies, which can give full play to the synergistic effect of surfactant reducing interfacial tension and polymer improving sweep volume to enhance oil recovery (Ge et al., 2021; Rai et al., 2013; Tan et al., 2022; Wei et al., 2022). In the 1960s, a series of SP flooding pilot tests were carried out by the United States and have achieved remarkable results (Dai et al., 2015; Shaker Shiran and Skauge, 2013). Subsequently, in the 1980s, development test areas were successively deployed in the Shengli, Xinjiang, and Daqing Oilfields in China, all of which have realized positive development results and economic advantages (Bai et al., 2015; Buijse et al., 2010; Chen et al., 2012). These results give substantial theoretical and technical support for decreasing water cut and increasing oil recovery in mature oilfields recovered by water injection (Ge et al., 2021). The EOR of SP flooding is expected to be 13% – 18% of the original oil in place (OOIP), between those of polymer flooding and ASP flooding (Zhu et al., 2012). The main technical features of SP flooding include five aspects: (1) great potential for EOR and a wide range of applicable reservoir conditions; (2) compared to ASP flooding, SP flooding avoids problems caused by alkali (Ding et al., 2019); (3) the ground injection process is simple, simplifying the injection process and anti-corrosion treatment (Zhu et al., 2012); (4) good economy, reducing the costs of chemical agents, equipment and operation (Zhu et al., 2012); (5) higher requirements for the surfactant, which makes it more challenging to achieve ultra-low interfacial tension under alkali-free conditions (Zhu et al., 2012).

The SP flooding technology has made a breakthrough in the industrial test of conglomerate reservoirs in Xinjiang Karamay Oilfield of China, which has matured supporting technologies (Liu et al., 2017; Tan et al., 2022). It has been identified as the central technology for greatly enhancing oil recovery in Xinjiang conglomerate reservoirs. Since 2015, large-scale SP flooding has been carried out in the Badaowan Formation of Well Block 530 and the Kexia Formation of Qixi Block, and a remarkable effect has been achieved (Tan et al., 2022). However, as the depth of the reservoir increases, the productivity footage ratio of the material decreases and the economic gain decreases. To realize efficient development of the conglomerate reservoir after water flooding and stimulation of the production-declined reservoir, dynamic regulation is necessary for the existing well pattern (Kim et al., 2021; Li et al., 2021; Zhan et al., 2022; Zhang et al., 2015, 2019; Zhao et al., 2018). Unlike the conventional five-spot well pattern, the inverted seven-spot well pattern has an injection-production well ratio of 1:2. Therefore, the injection fluid has a larger sweep area but a weaker sweep intensity. This characteristic of the inverted seven-spot well pattern will inevitably affect the application potential of the SP system. It is vital to evaluate the adaptability and the development benefit of the SP system in an inverted seven-spot well pattern.

Five oil layers, J_1b_1 , J_1b_2 , J_1b_3 , J_1b_4 , and J_1b_5 , are developed from top to bottom in the Badaowan Formation, with an average permeability of $173.91 \times 10^{-3} \mu\text{m}^2$. The target horizons for chemical flooding are J_1b_4 and J_1b_5 (abbreviated as J_1b_{4+5}). Its lithology is distributed in two sections: the upper is the sandstone section and the lower is the conglomerate section. Based on the reservoir physical properties of J_1b_{4+5} , different indoor models were established in this paper. First, the pressure and pressure gradient distributions of two well patterns were analysed by numerical simulation, and the different swept intensity regions were divided. Second, by regulating the displacement rate, the single-core displacement experiment was used to obtain the ability of the SP system to recover crude oil under different swept intensities. The comparison of the EOR potential in the two well patterns was obtained on this basis. Thirdly, the migration law and the oil displacement efficiency of the SP system under the sand-gravel mixture condition were clarified by the heterogeneous cuboid core displacement experiments. Finally, the migration law of injection fluid in the inverted seven-spot well pattern and the comparison of EOR capacity under different SP system injection strategies were clarified using a one-third inverted seven-spot configuration. After determining the appropriate injection approach, the EOR capacities in two well patterns were compared in depth.

Experimental

Materials

Hydrophobic associated polymer KYPAM with a relative molecular mass of 1.62×10^7 and a hydrolysis degree of 26.76% was supplied by Beijing Hengju Chemical Group Co., Ltd. Naphthenic petroleum sulfonate KPS with a molecular weight of 452 g/mol and an active content of 30.05 wt.% was provided by China National Petroleum Corporation Xinjiang Oilfield Branch.

The dehydrated crude oil was obtained from Karamay Oilfield, China, with a density of $0.8763 \text{ g}\cdot\text{cm}^{-3}$ and a viscosity of $39.54 \text{ mPa}\cdot\text{s}$ at the reservoir temperature (35.6°C). The simulated formation water with total dissolved solids (TDS) of $14,466.24 \text{ mg}\cdot\text{L}^{-1}$ was prepared according to the ion composition of natural formation water in Karamay Oilfield, shown in Table 1.

Table 1—Karamay Oilfield simulated formation water.

Composition	$\text{K}^+\text{+Na}^+$	Mg^{2+}	Ca^{2+}	Cl^-	SO_4^{2-}	HCO_3^-	TDS	pH
Concentration ($\text{mg}\cdot\text{L}^{-1}$)	5145.66	27.04	208.06	7388.7	22.26	1674.52	14,466.24	7.45

Comparison of seepage intensity under different well pattern conditions

In order to clarify the influence of well pattern type on the seepage intensity and obtain the distribution law of the seepage intensity, tNavigator was used to simulate the well patterns (the inverted seven-spot well pattern and the five-spot one) under the condition of single-phase flow, and MATLAB was used for post-processing. For comparison, the producer-injector spacing was set to 150 m. The injection well was injected quantitatively, while the production wells maintained constant pressure. The obtained pressure distributions are depicted in Fig. 1.

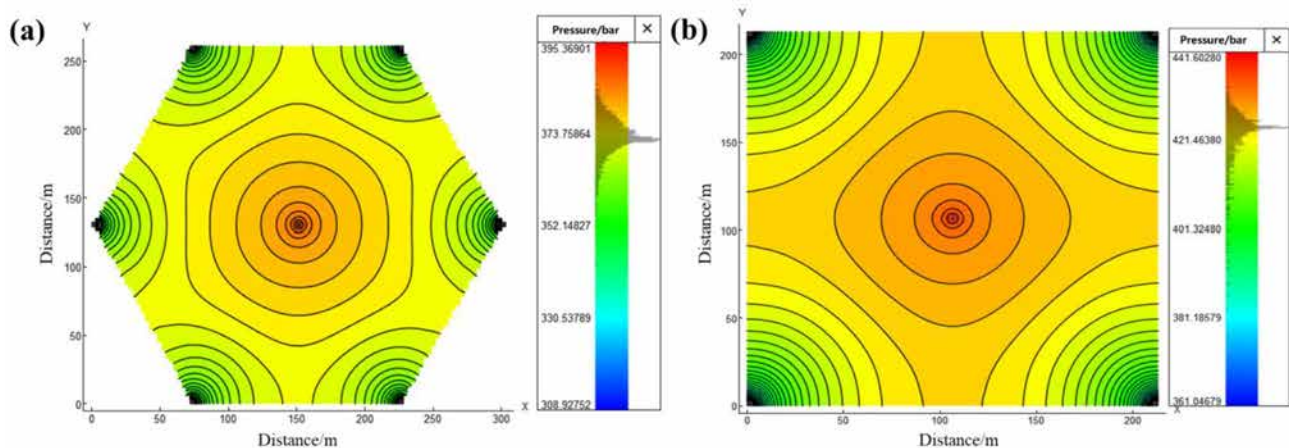


Figure 1—Pressure distributions of (a) inverted seven-spot and (b) five-spot well patterns.

The central difference method was employed to calculate the pressure gradient $\left(\frac{\partial P}{\partial x} \text{ and } \frac{\partial P}{\partial y}\right)$ distribution in the X and Y directions once the pressure data had been obtained. The solutions for the inner grid (m, n) and the boundary grid (M, N) of the well patterns are shown in Eqs. (1) and (2), respectively. The corresponding diagrams are shown in Fig. 2.

$$\begin{aligned} \frac{\partial P}{\partial x} \bigg|_{(m,n)} &= \frac{P(m+1,n) - P(m-1,n)}{2 \times \Delta x} \\ \frac{\partial P}{\partial y} \bigg|_{(m,n)} &= \frac{P(m,n+1) - P(m,n-1)}{2 \times \Delta y} \end{aligned} \quad (1)$$

Design and fabrication of cuboid and flat cores

In order to clarify the channeling law of the injection fluid in the sand-gravel mixture reservoir and the EOR capability of different injection methods, the sand-gravel mixture cuboid core (length 30 cm $\times$ width 4.5 cm $\times$ thickness 4.5 cm, average permeability $154 \times 10^{-3} \mu\text{m}^2$, average porosity 21.7%) shown in Fig. 3 was designed. The core was formed by superimposing the upper sandstone layer ($300 \times 10^{-3} \mu\text{m}^2$) and the lower conglomerate layer ($50 \times 10^{-3} \mu\text{m}^2$) of equal thickness (2.25 cm), and its surface was formed by pouring epoxy resin. The core included an injection port, two production ports, and six pressure measurement taps. The injection port was shared by the two layers, while the two production ports were set to correspond to the sandstone layer and the conglomerate layer, respectively. The precise placements of six pressure taps on both sides of the core are shown in Fig. 3.

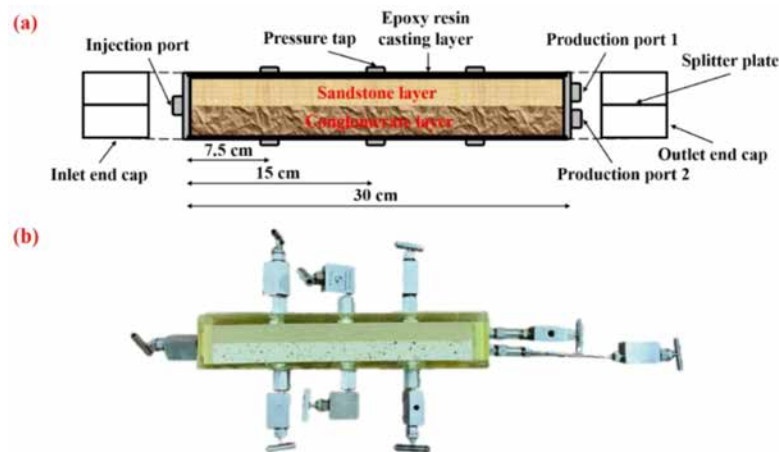


Figure 3—Sand-gravel mixture cuboid core based on commingled injection and separated production technology. (a) the schematic and (b) the actual picture.

A one-third inverted seven-spot configuration, sketched in Fig. 4a, was designed to investigate the migration law of injection fluid in the inverted seven-spot well pattern and the oil displacement impact under various injection strategies. According to the streamline distribution and the injection-production well ratio in the well pattern, the designed core (the corresponding regular hexagon had a side length of 24 cm) was cut from a flat core (length 30 cm $\times$ width 30 cm $\times$ thickness 4.5 cm, permeability $300 \times 10^{-3} \mu\text{m}^2$, porosity 24.1%). An injection well, two production wells, and several electrodes were arranged on the core according to the positions shown in Fig. 4a. The electrodes were used to monitor the change of electrical resistance in each position in real time. An inversion calculation from the resistivity-saturation standard curve was adopted to reflect the oil saturation distribution at different stages. To further compare the differences in the migration law and the EOR capability of injection fluid in different well pattern types, a one-fourth five-spot configuration (permeability $300 \times 10^{-3} \mu\text{m}^2$, porosity 24.1%) with one injection well, one production well, and several electrodes was designed and fabricated according to the same idea, as shown in Fig. 4b. According to the principle of equal control reserves per well, the length, width, and thickness of the one-fourth five-spot configuration were 19.5 cm, 19.5 cm, and 4.5 cm respectively.

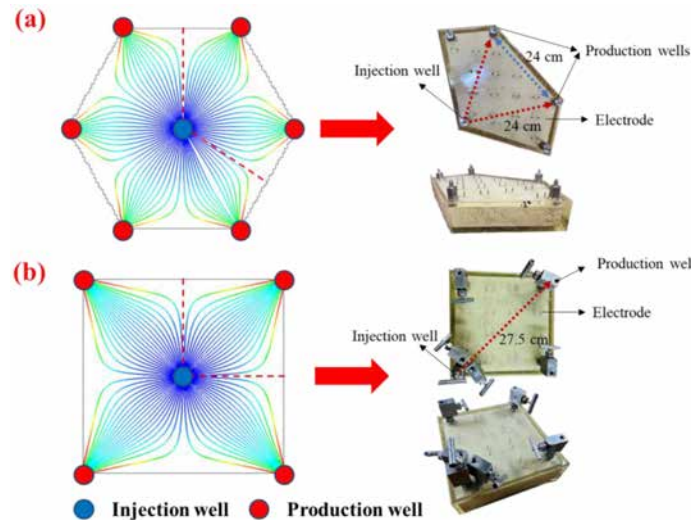


Figure 4—Design of (a) inverted seven-spot and (b) five-spot well pattern cores.

Migration law of SP system in the sand-gravel mixture condition

The experimental apparatuses for this part are shown in Fig. 5a. The temperature and fluid injection rate were maintained at 35.6°C and $0.3\text{ mL}\cdot\text{min}^{-1}$, respectively. The brine and SP system solutions utilized here were identical to those in single-core displacement tests. The core was vacuumed and saturated with simulated water prior to the experiment. The precise experimental procedure consisted of the following steps: (1) inject the simulated water until the volume of fluid produced per unit time in the high-permeability layer and the low-permeability layer is constant; (2) inject 1.0 PV of SP system solution; (3) perform the subsequent water flooding until the fluid diversion ratios (ratio of instantaneous liquid production of each layer to total instantaneous liquid production) of the sandstone and conglomerate layers at the outlet tend to be stable. Variation in pressure and the produced fluid volume were recorded throughout the test to evaluate the migration law of the SP system under the conditions of the sand-gravel mixture.

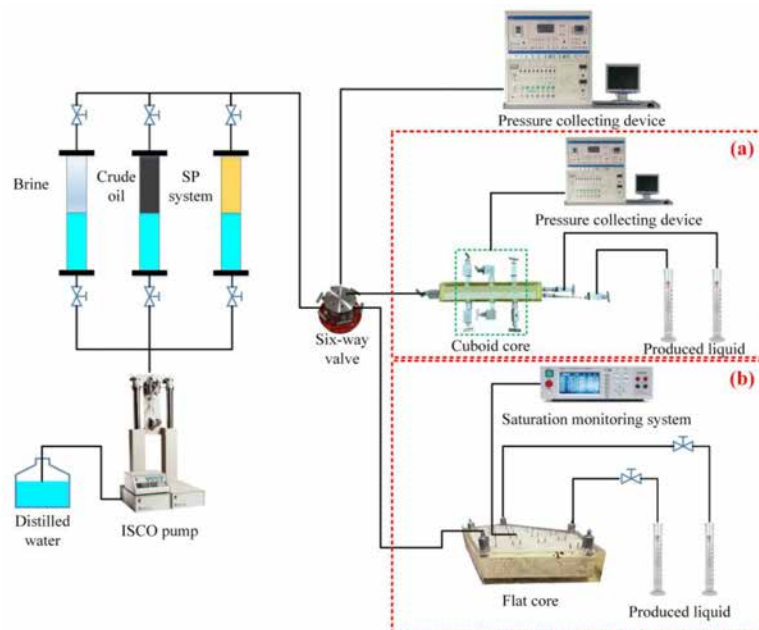


Figure 5—Diagram of the percolation and oil displacement experiments in (a) the cuboid core and (b) the flat core.

EOR capacity of SP system under different injection methods based on the heterogeneous model

The experimental setups and conditions for this part were the same as those in the previous section. The brine, SP system solution (constant 0.3 wt.% KPS and varying concentrations of KYPAM), and crude oil were utilized here. For comparability of results, the total amount of polymer used for SP slugs in each set of experiments was almost equal, as shown in Table 3. The viscosities of the three SP system slugs from low to high were: 20.2, 39.8, and 54.7 mPa·s. To establish the initial conditions, the cuboid cores were vacuumized and saturated with the simulated water. The core was then saturated with the dehydrated oil at a rate of 0.1 mL·min⁻¹ and aged for two days. After ensuring that the core reached the initial oil saturation ($S_{oi}=56.5\%$), the water flooding test started. Once the maximum oil recovery with water flooding was achieved (the water cut was over 98%), 0.9 PV of SP slug was injected, followed by the subsequent water flooding to accomplish the ultimate oil recovery. The injection pressure differential was recorded throughout, and the oil recovery at each stage was calculated to compare the difference in the EOR capability of the different injection strategies.

Table 3—Schemes of the EOR capacity experiments under different injection methods based on the heterogeneous model

No.	Injection strategy	SP system slug design	
		KPS concentration	KYPAM concentration, slug size
1	Constant viscosity injection	0.3 wt.%	2200 mg·L ⁻¹ (0.9 PV)
2	Step-up viscosity injection	0.3 wt.%	1500 mg·L ⁻¹ (0.2 PV) + 2200 mg·L ⁻¹ (0.5 PV) + 2700 mg·L ⁻¹ (0.2 PV)
3	Step-down viscosity injection	0.3 wt.%	2700 mg·L ⁻¹ (0.2 PV) + 2200 mg·L ⁻¹ (0.5 PV) + 1500 mg·L ⁻¹ (0.2 PV)

Migration characteristic and EOR potential of SP system in the inverted seven-spot well pattern

This section sought to investigate the SP system's migration characteristics and EOR potential in the inverted seven-spot well pattern. Fig. 5b illustrates the flat core flooding and data acquisition systems. The temperature was kept at 35.6°C, and the injection rate at 0.3 mL·min⁻¹. The SP system slug design was the same as stated in Table 3, based on which 0.9 PV of low-viscosity system (0.3 wt.% KPS and 1500 mg·L⁻¹ KYPAM) constant viscosity injection test was also carried out. Before each experiment, the core was vacuumized and fully saturated with brine. The core was then flooded with filtered crude oil until brine production stopped and aged for two days. The S_{oi} of the flat core was about 70%. The specific experimental process was as follows: (1) inject the simulated water until maximum water flooding recovery is achieved; (2) inject the SP system solution; (3) perform the subsequent water flooding until the water cut is greater than 98%. During the experiments, liquid production, injection pressure, and electrical resistance were measured. The optimal SP system injection strategy was obtained by comparing the EOR values. A similar methodology was adopted in the five-spot well pattern core to compare the migration law and oil displacement efficiency of the SP system under different well patterns using the obtained optimal strategy.

Results and Discussion

Seepage intensity division under different well patterns

According to the pressure gradient distribution (Fig. 6a), there are seven high-pressure and six low-pressure gradient zones in the inverted seven-spot well pattern. The centers of high-pressure gradient zones are injection and production wells, while those of low-pressure gradient zones are located at the equilibrium points on the connection line of adjacent oil wells. Similarly, the five-spot one has five high-pressure and four low-pressure gradient zones, respectively. Referring to the oilfield development data, the swept intensity of different regions in the well pattern is divided according to the pressure gradient. The specific division basis is detailed in Table 4. It can be seen from Fig. 6b that the strong swept regions are almost distributed in the bottom hole and surroundings; the medium swept regions of the seven-spot well pattern

are distributed in the near-well zone, while those of the five-spot one are mainly distributed on the producer-injector connection line; the weak swept regions of the seven-spot well pattern are distributed in the far-well zone, while those of the five-spot one are near the connecting line of the production wells; in addition, the ineffective swept regions are all distributed on the connecting line of the production wells. By comparing the pie chart of the swept intensity classification (Fig. 6c), it can be found that the inverted seven-spot well pattern presents a weak swept state overall, and the weak swept region accounts for 61%, which is almost twice that for the five-spot one. The difference in seepage intensity is bound to be reflected in the EOR capacity of the SP system, necessitating a study into the system's adaptation to different well patterns.

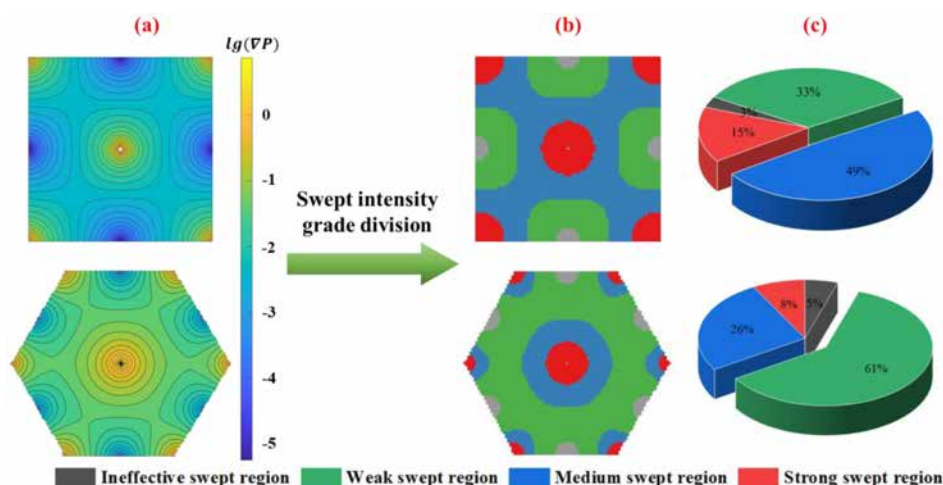


Figure 6—Difference of two well patterns in (a) pressure gradient distribution, (b) swept intensity region distribution, and (c) different swept intensity region proportion.

Table 4—Classification standard of different swept regions.

Classification	Strong swept region	Medium swept region	Weak swept region	Ineffective swept region
Criterion	> 0.2 MPa/m	0.1 – 0.2 MPa/m	0.03 – 0.1 MPa/m	< 0.03 MPa/m

Influence of seepage intensity on EOR efficiency

In this section, the crude oil recovery capacity of the SP system under varied seepage intensities was obtained by single-core experiments, and the difference in EOR potential in different well patterns was given in combination with the results of the previous section. Different injection rates were used to represent the seepage intensities in different regions of the well patterns. $0.3 \text{ mL} \cdot \text{min}^{-1}$ represented the strong seepage intensity, $0.1 \text{ mL} \cdot \text{min}^{-1}$ represented the medium, and $0.05 \text{ mL} \cdot \text{min}^{-1}$ represented the weak. Fig. 7 listed the EOR capacities of the two SP systems (20.2 and 39.8 mPa·s, respectively) under different seepage intensities. The EOR capacity of the high-viscosity system was stronger than that of the low-viscosity one under the same conditions. Furthermore, regardless of the high- or low-viscosity systems, with the increase in seepage intensity, the EOR capacity was significantly improved, and the increased range was 15% to 20%. Furthermore, by comparing the EOR ability and injection pressure under two permeability conditions, it can be found that the SP system had better performance under the high permeability condition, and the corresponding injection pressure was lower, indicating that the system was easier to penetrate the high-permeability layer.

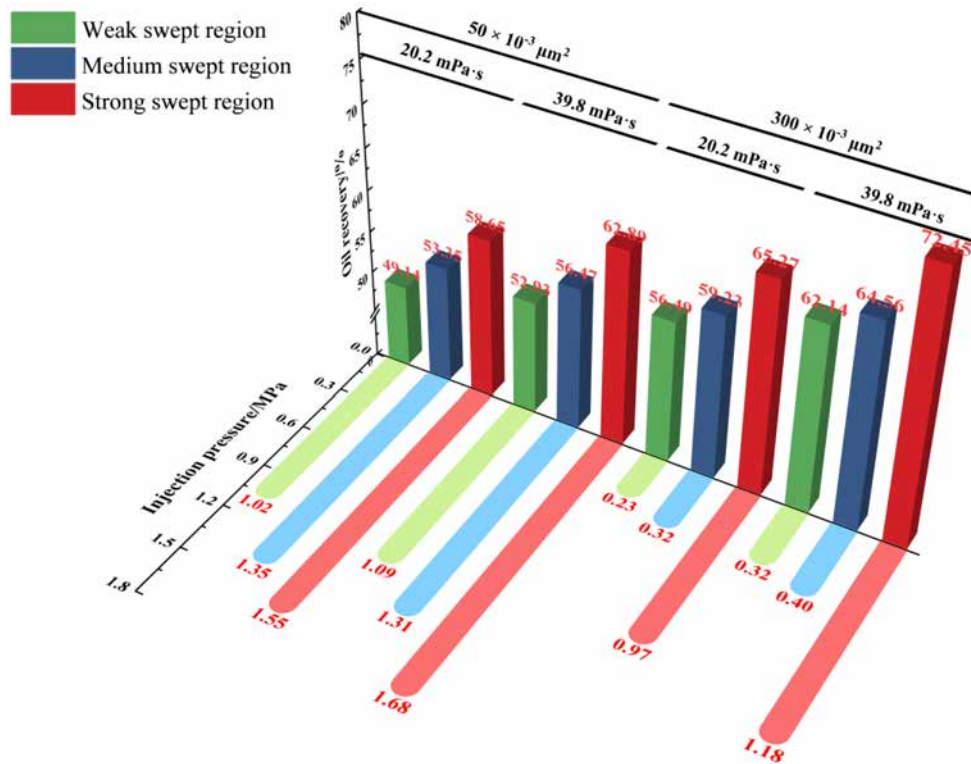


Figure 7—Difference in EOR capacities of two SP systems under different seepage intensities.

Concerning the whole well pattern, the difference in EOR capacity induced by varying swept intensities will inevitably be reflected in the EOR potential of the SP system for different well patterns. The data in Fig. 7 are weighted and averaged according to the different swept intensity region proportions divided in Fig. 6c to obtain the EOR potential of the SP system in the whole well patterns. The results are shown in Table 5. It is not difficult to find that due to the dominance of the weak swept region in the inverted seven-spot well pattern, the EOR potential of the SP system in it is lower than that in the five-spot one, and the difference is about 4 percentage points.

Table 5—Calculating EOR potential of SP systems in two well patterns.

Permeability/ $\times 10^{-3} \mu\text{m}^2$	SP system Viscosity/ mPa·s	Calculating EOR potential/%		
		In inverted seven-spot well pattern	In five-spot well pattern	Difference
50	20.2	48.54	52.87	-4.33
50	39.8	52.00	56.36	-4.36
300	20.2	55.08	59.04	-3.96
300	39.8	60.49	64.86	-4.37

Migration law in the heterogeneous model

In order to explore the conformance control ability of SP systems in the heterogeneous reservoir, Fig. 8 illustrates the variation of the liquid diversion ratio in sandstone and conglomerate layers before and after injecting SP slugs. Before taking measures, the sandstone layer was the absolute dominant channel for liquid flow with a diversion ratio of as high as 96%, which was changed after the injection of SP systems. In the sand-gravel mixture reservoir, the SP system would preferentially enter the high-permeability sandstone layer and play a certain conformance control role, resulting in the subsequent injection fluid turning to the low-permeability conglomerate layer, expanding the sweeping range of the displacement phase. As shown

in Fig. 8a, injecting the low-viscosity system had minimal influence on the diversion ratio at the early stage. After the injection of 0.8 PV of low-viscosity slug, the diversion ratio of the sandstone layer decreased obviously, reaching a minimum of 56.79% at 0.87 PV of slug. Then the sandstone layer's diversion ratio rose quickly to 95% after subsequent water flooding. It was almost the same as before the measure was taken, indicating that the low-viscosity system has a limited ability to improve the sweeping range of the displacement phase in the sand-gravel mixture condition. From Fig. 8b, after injection of the high-viscosity system, the diversion ratio changed significantly and immediately. After injecting 0.73 PV of SP system, the diversion ratio of the sandstone layer and conglomerate layer reversed, and the diversion ratio was 47.37%:52.63%. The rising rate of the diversion ratio of the sandstone layer and its ultimate diversion ratio after the subsequent water flooding were significantly lower than those of the low-viscosity system. The ultimate diversion ratio was 89.5%, 6.5 percentage points lower than the initial. The results indicate that the high-viscosity system is more suitable for the heterogeneous sand-gravel mixture reservoir. The injection volume should be greater than 0.73 PV to better utilize the remaining oil in the conglomerate layer.

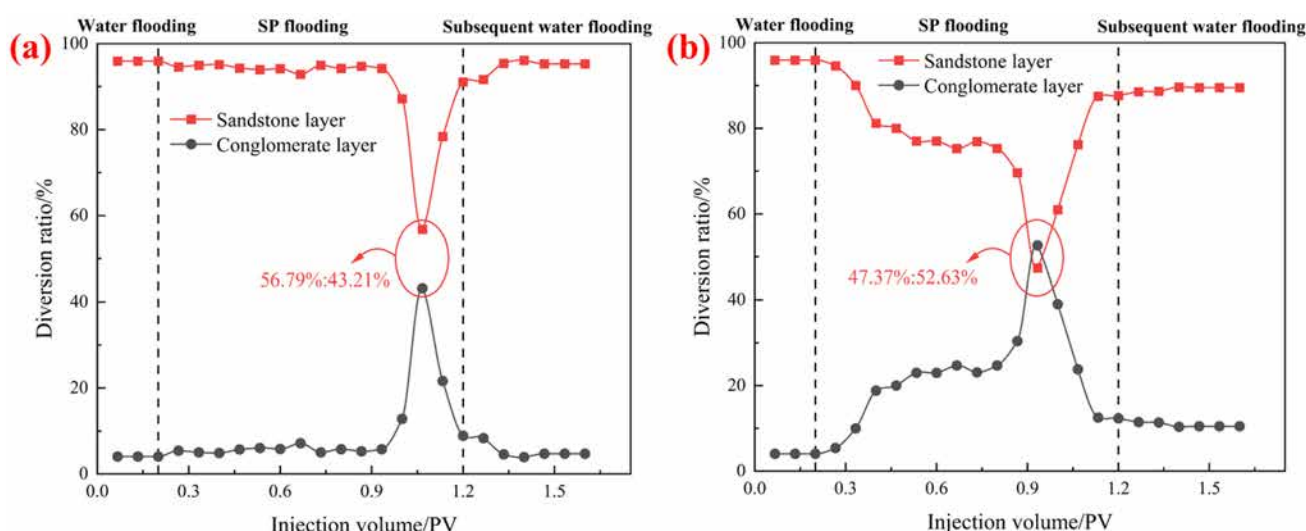


Figure 8—Variation of diversion ratio of the heterogeneous model before and after injection of (a) low-viscosity (20.2 mPa·s) and (b) high-viscosity (39.8 mPa·s) SP systems.

Moreover, as shown in Fig. 9, by comparing the pressure variations along the path, it is clear that the two systems occupy distinct acting locations inside the sand-gravel mixture cuboid cores. With the injection of the low-viscosity system, the pressure of sandstone layers 1 and 2 gradually began to rise, followed by those of sandstone layer 3 and conglomerate layer 2. The order of final stable pressure was: sandstone layer 1 > sandstone layer 2 > conglomerate layer 2 > sandstone layer 3. Furthermore, the pressure of conglomerate layers 1 and 3 changed little throughout the whole process. It illustrates that with the flushing of the liquid flow, the SP system mainly accumulates in the middle of the sandstone layer, and the fluid diversion occurs only in the middle of the formation. Regarding the injection of the high viscosity system, the pressure of conglomerate layers 1 and 2 increased immediately after the pressure of the sandstone layer increased. The final stable pressure of conglomerate layer 1 was slightly higher than that of sandstone layer 2, while the pressure of conglomerate layer 2 was between that of sandstone layers 2 and 3. It shows that compared with the low-viscosity system, the high-viscosity one has a stronger conformance control ability, which can accumulate in the front and middle of the sandstone layer and effectively recover the remaining oil in the conglomerate layer at the corresponding position.

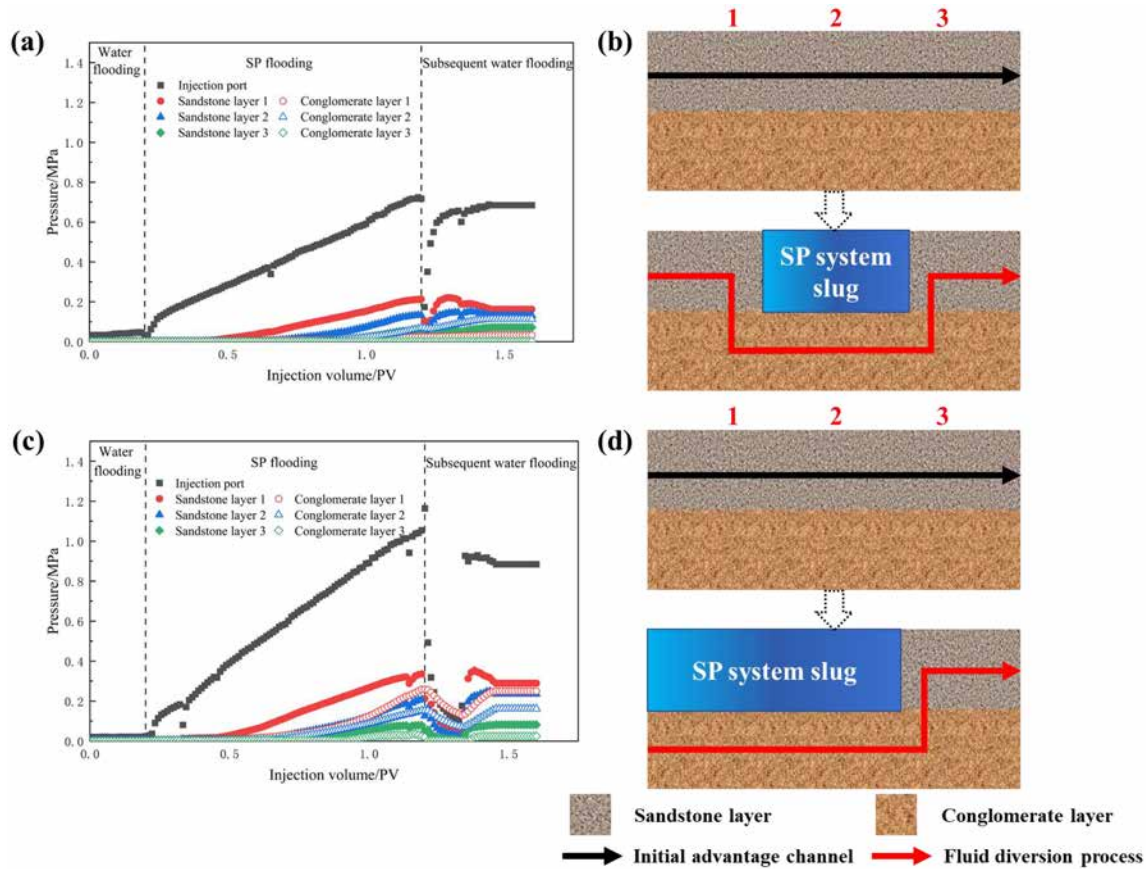


Figure 9—Pressure variations before and after injection of (a) low- and (c) high-viscosity systems; schematic diagrams of flow diversion caused by (b) low- (20.2 mPa·s) and (d) high-viscosity (39.8 mPa·s) systems.

Optimization of injection methods based on the heterogeneous model

In the previous section, it can be seen that appropriately increasing the viscosity of the SP system can improve its conformance control ability under heterogeneous conditions. In this section, the influence of different injection methods on oil recovery performance was evaluated. Figs. 10a to 10c show the dynamic characteristic curves of oil displacement under three different SP system injection methods in the heterogeneous cores. For each case, the water flooding stage lasted about 0.6 PV, and the corresponding oil recovery rate was only about 35%. However, the curves showed significant differences after the injection of 0.9 PV of SP systems under different injection strategies. Under the step-down viscosity injection strategy (Fig. 10c), the comprehensive water cut of the produced fluid decreased the fastest and the lowest water cut could be obtained compared with the other strategies. Meanwhile, the resulting increase in injection pressure and the final stable pressure were also the largest among the three injection methods. It is mainly because under the step-down viscosity injection strategy, the preferentially injected high-viscosity slug results in an immense pressure gradient of the displacement front, which can form a highly concentrated oil bank, substantially minimize the water cut, and obviously increase the flow resistance in the sandstone layer. Therefore, subsequent medium- and low-viscosity slugs are easier to enter the conglomerate layer, which considerably increases the swept volume of the conglomerate layer and ameliorates the mobility difference between the two layers. Correspondingly, the EOR capacity in the experiments also showed a corresponding gap (Fig. 10d). The EOR value for the constant viscosity injection was 18.82%, for the step-up viscosity injection was 17.03%, and for the step-down viscosity injection was 25.47%.

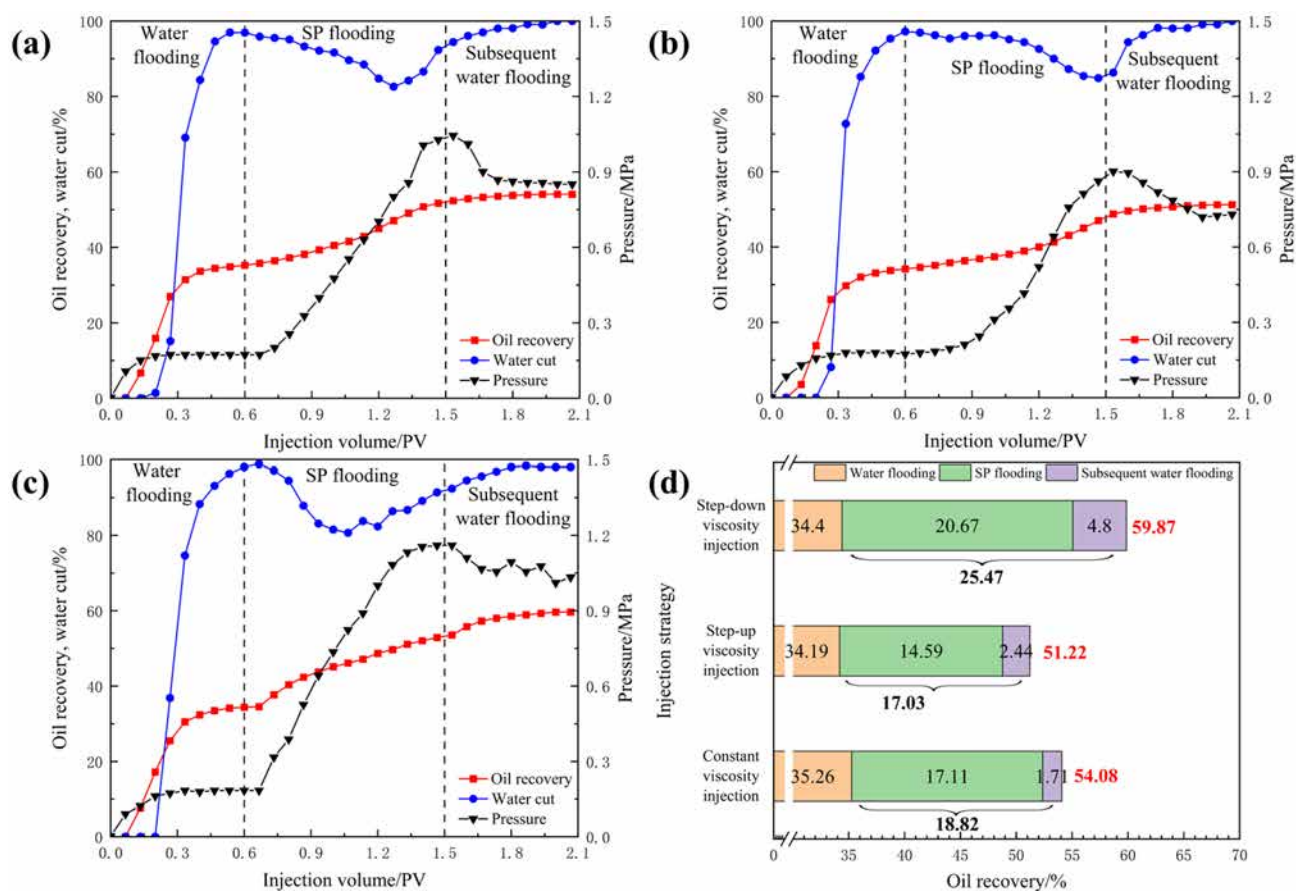


Figure 10—Dynamic production curves under different SP system injection methods in the heterogeneous cores: (a) constant viscosity injection, (b) step-up viscosity injection, (c) step-down viscosity injection, and (d) EOR amplitude at each stage.

Optimization of injection methods and EOR ability in different well patterns

First, the sweep ability of SP systems with different viscosities in the inverted seven-spot well pattern was studied. Fig. 11 lists the oil saturation distribution of the inverted seven-spot well pattern before and after displacement by SP systems under 0.9 PV constant viscosity injection. The average oil saturation $\bar{S}_o$ after water flooding (0.933 PV) was about 36% (36.1% in Fig. 11a and 35.9% in Fig. 11b). Furthermore, the oil saturation on the connecting line of injection-production wells was relatively low, indicating that a certain dominant channel was formed after water flooding. As shown in Fig. 11a, the low-viscosity system could inhibit water channeling to a certain extent, so the effective affected area increased. However, the accumulation degree of crude oil in the dead oil area reminded high. The waterline broke through rapidly when undertaking subsequent water flooding and could not further affect the non-primary streamline area. The $\bar{S}_o$ of the well pattern decreased to 26.4%, a loss of 9.6 percentage points. Compared with the low-viscosity system, the high-viscosity system could more effectively restrain waterline breakthrough, increasing the effective sweep area and decreasing the crude oil saturation in the dead oil zone substantially (Fig. 11b). In the subsequent stage of water flooding, the water breakthrough was delayed, which further reduced the oil saturation in the non-primary streamline area. The $\bar{S}_o$ was 24.3% and 2.1 percentage points lower than that treated by the low-viscosity system. It indicates that the viscosity of the SP system applied to the inverted seven-spot well pattern should be appropriately increased under the premise of satisfying the formation compatibility.

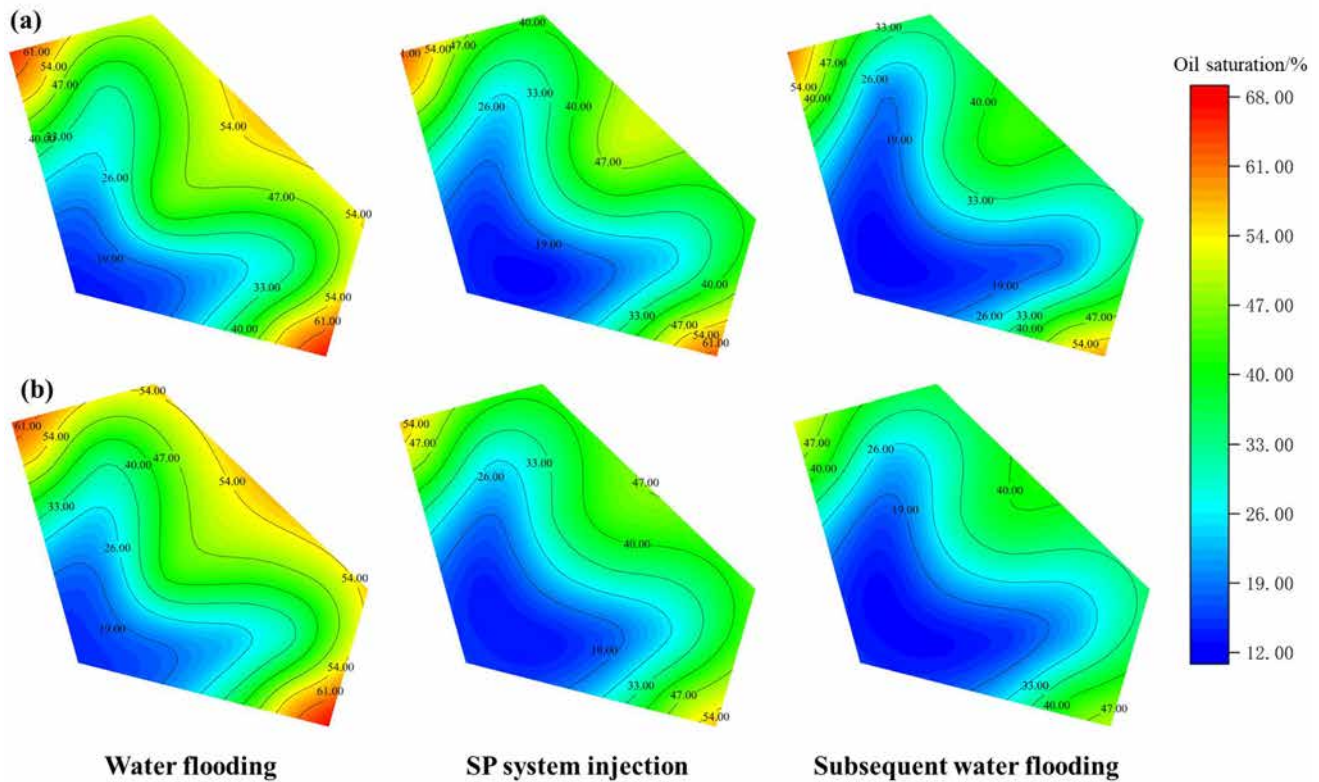


Figure 11—Oil saturation distribution of inverted seven-spot well pattern before and after displacement by (a) low- (20.2 mPa·s) and (b) high-viscosity (39.8 mPa·s) systems under 0.9 PV constant viscosity injection.

Then, the influence of injection methods on the EOR ability of SP systems in the inverted seven-spot well pattern was evaluated. Figs. 12a to 12c show the dynamic production curves of the inverted seven-spot well pattern model under three different 0.9 PV of SP system injection strategies. It can be found that under experimental conditions, the water-free stage of the inverted seven-spot well pattern was 0.133 PV, and the corresponding recovery rate was about 21%. After 0.933 PV of water flooding, the water cut reached 95%, and the water flooding recovery was about 44.50%. Subsequently, various SP system injection strategies were applied to the model (the same as stated in Table 3). The EOR capacities from high to low were: step-down viscosity injection (22.02%) > constant viscosity injection (18.05%) > step-up viscosity injection (15.90%). In conjunction with the water content variation curve, it was found that the decreased range of water cut and the appearance time of the lowest water cut were positively correlated with the EOR capacity. The phenomenon indicates that the high-viscosity slug injected preferentially can form a highly concentrated oil bank at the displacing front, which has a stronger mobility control ability than the low-viscosity one, reducing the water cut quickly and efficiently, and increasing the sweeping area of the injected fluid in the well pattern. Therefore, under the condition of a fixed slug size in the SP system, step-down viscosity injection should be preferred to further enhance oil recovery.

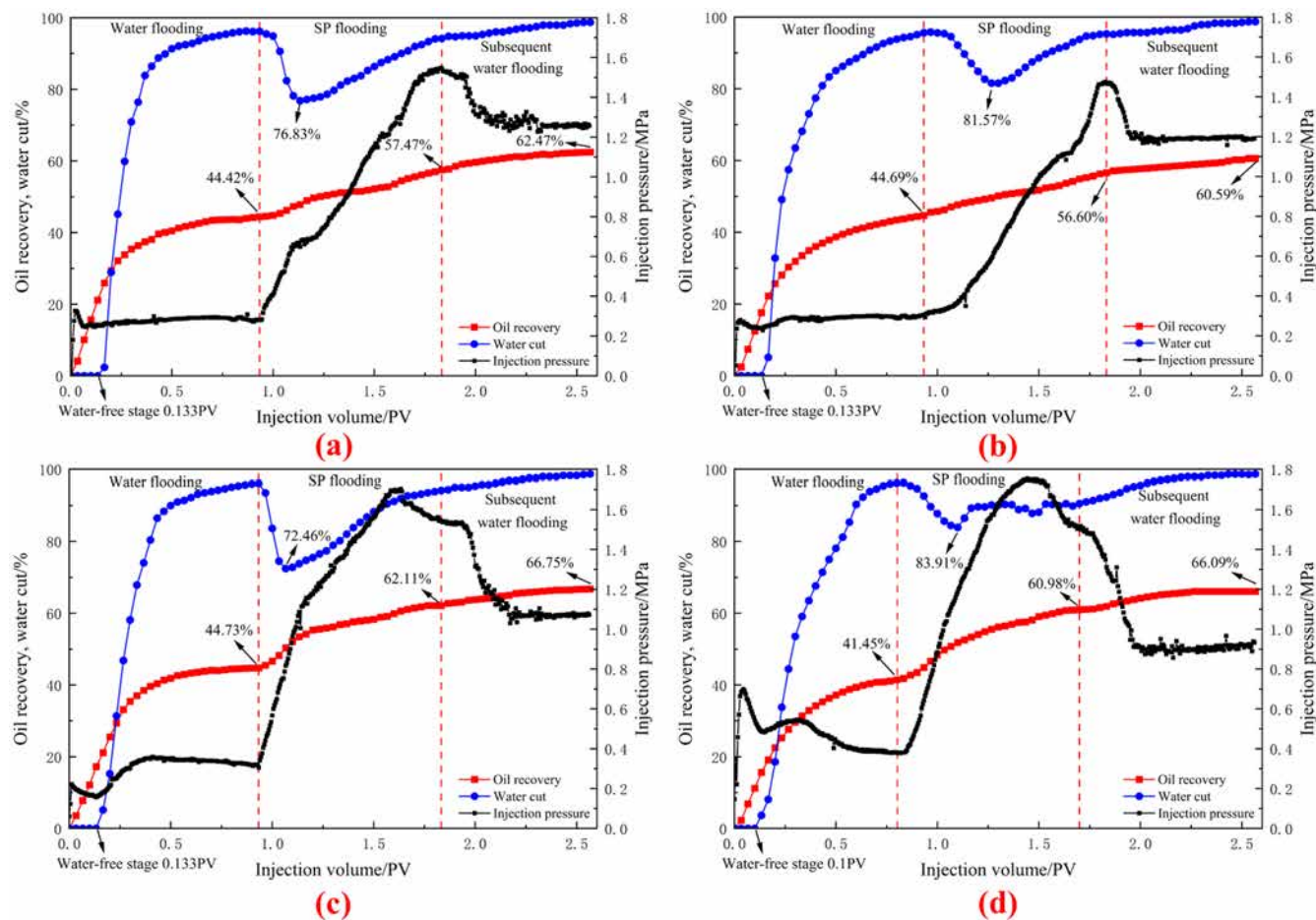


Figure 12—Dynamic production curves under different SP system injection methods in the inverted seven-spot well pattern: (a) constant viscosity injection, (b) step-up viscosity injection, and (c) step-down viscosity injection; (d) dynamic production curves under step-down viscosity injection in the five-spot well pattern with equal control reserves.

For further comparison of the fluid migration law and the EOR ability of the SP system in different well patterns, the dynamic production curve and the oil saturation distribution under the optimal injection method (step-down viscosity injection) were listed in Figs. 12c, 12d, and 13. The water-free stage of the five-spot well pattern was 0.1 PV, and the corresponding oil recovery was about 11%. After 0.8 PV of water flooding, water flooding recovery reached 41.45%. Compared to the inverted seven-spot well pattern, the stage of water flooding was shorter, and the water flooding recovery rate was lower. In addition, as shown in Fig. 13, the advantageous channel for injection fluid in the five-spot well pattern was more apparent, and the crude oil in dead oil zones could hardly be recovered ($S_o > 68\%$ in these zones). After SP system injection and subsequent water flooding, the ultimate oil recovery of the five-spot well pattern was 66.09%, almost the same as that of the inverted seven-spot one. The EOR was 24.64% and 2.62 percentage points higher than that in the inverted seven-spot one. Apparent differences can be found in the sweeping area and oil washing capacity of the SP system in the two well patterns (Fig. 13). In order to compare more intuitively, the oil saturation before and after the displacement of the SP system was subtracted, and the obtained value was divided by the pre-displacement saturation to acquire the percentage of oil saturation reduction in each part of the well pattern, as shown in Fig. 14. The SP system in the inverted seven-spot well pattern had a wide range of spread, including the middle of the model, the primary streamline area, and the dead oil area. However, the proportion of high-efficiency swept area was small, resulting in the relatively low EOR capacity of the SP system in the inverted seven-spot well pattern. Compared with the inverted seven-spot well pattern, the swept area and the producing degree of the SP system in the five-spot one were more and stronger concentrated in the primary streamline area, while the corner area was less utilized. According to

the threshold of reducing oil saturation by more than 20%, the effective swept areas of the inverted seven-spot well pattern and the five-spot one were 93.3% and 68.0%, respectively.

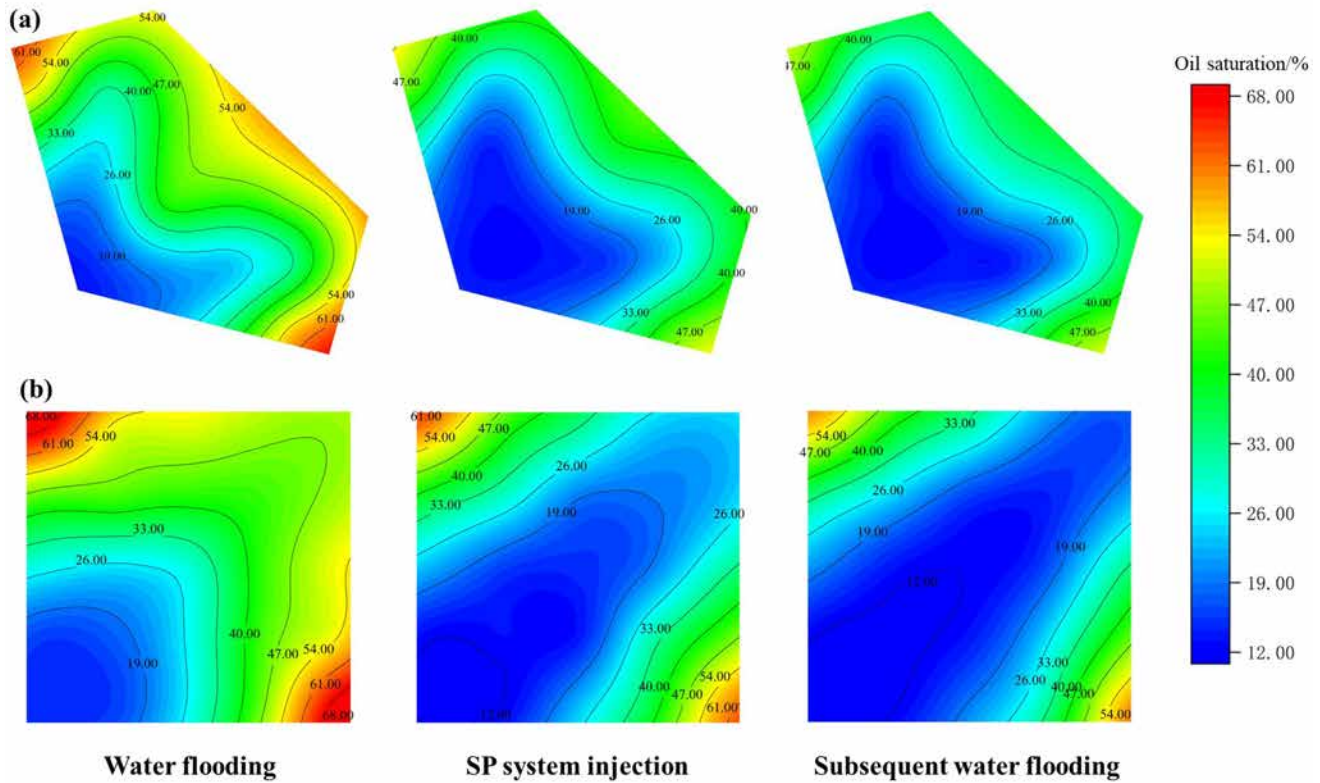


Figure 13—Oil saturation distribution of (a) inverted seven-spot and (b) five-spot well patterns after different stages under the optimal SP system injection method.

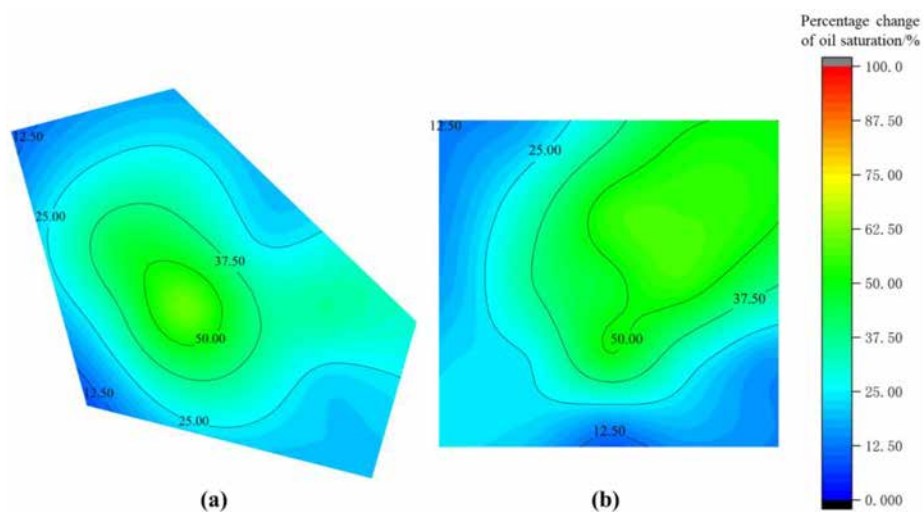


Figure 14—Percentage change of oil saturation in (a) inverted seven-spot and (b) five-spot well patterns under optimal injection method.

In general, compared to the five-spot well pattern, the swept area of the SP system in the inverted seven-spot well pattern was larger, but the intensity was weaker. When converting well pattern types, injection and production wells should be reasonably arranged so that the remaining oil and residual oil that is not recovered in the five-spot well pattern can be efficiently recovered by utilizing the characteristics of the inverted seven-spot well pattern.

Conclusions

In this research, numerical simulation was used to determine the distributions of distinct seepage intensity zones in two well patterns. EOR abilities of the SP system under different seepage intensities were obtained by single-core oil displacement experiments. Heterogeneous cuboid cores were used to simulate the sand-gravel mixture condition, and the seepage law and the displacement ability of the SP system were obtained. The sweeping capacity and the oil displacement efficiency under different SP system injection strategies were compared using the custom-made flat cores, and the adaptability of the SP system in two well patterns was evaluated. The following are the particular conclusions:

1. Compared with the five-spot well pattern, the overall pressure gradient of the inverted seven-spot well pattern is significantly lower, showing a low intensity swept state. The weak swept region accounts for 61%.
2. With the increase in seepage intensity, the oil-producing capacity of the SP system increases by about 15% – 20 %. Increasing the reservoir permeability and viscosity of the SP system can improve the oil displacement effect in each swept region.
3. To further enhance the utilization degree of the conglomerate layer, the viscosity and slug size of the SP system should be adequately increased within a reasonable range, and the step-down viscosity injection strategy should be adopted.
4. Step-down viscosity injection strategy can obtain the maximum swept area and EOR rate in the inverted seven-spot well pattern. In comparison to conventional constant viscosity injection, the recovery value increases by 3.97 percentage points.
5. The effective swept area of the SP system in the seven-spot well pattern (93.3%) is larger than that in the five-spot one (68.0%). The SP system in the seven-spot well pattern has better oil recovery capacity in the non-primary streamline area. However, the capacity in the primary streamline area is inferior to that in the five-spot well pattern. There is minimal variation between the two well types' ultimate oil recovery.
6. In addition to providing technical support for the design of injection parameters for SP flooding in Karamay conglomerate reservoirs, the above findings may serve as a guide for the popularization and application of SP flooding technology in the inverted seven-spot well pattern.

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